

Ruoxi (Anna) Shang

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Research Summary

I conduct research at the intersection of HCI and AI. My work focuses on **evaluating and steering LLM behavior** for trust, alignment, and reliability. I develop theory-grounded **benchmarks and evaluation pipelines** to measure model risks in realistic settings, and I design human-in-the-loop experiments to audit the downstream impact of AI agents. I have published 10+ peer-reviewed papers (CHI, EMNLP, FAccT, AIES) and contributed to foundational research at Microsoft Research and Oumi AI.

Education

University of Washington, Human-Centered Design and Engineering

Sep 2020 – Exp. Oct 2026

Ph.D. Candidate, Committee: Gary Hsieh, Chirag Shah, Tim Althoff, David McDonald

University of California, Berkeley

Graduated May 2020

B.A., Applied Mathematics and Statistics (Concentration: Data Science)

Skills

Programming: Python, JavaScript/TypeScript, React, PyTorch, scikit-learn, SQL, HTML/CSS

LLMs: Prompt Engineering, In-Context Learning, RAG, LLM APIs, Multi-agent frameworks, LLM-as-a-judge, Evaluation and Benchmark Design (e.g. taxonomy development, ablation and robustness test, human annotation)

HCI: Experimental design, Hypothesis testing, Mixed-methods, Survey Design, Think-aloud protocol, Affinity diagramming, Cognitive walkthrough, Heuristic evaluation, Rapid prototyping

Research & Work Experience

Oumi AI

Oct 2025 – Present

Research Fellow

Collaborators: Emmanouil Koukoumidis; Panos Achlioptas; Shang Hong Sim

- Building a **benchmark for semantic risk reasoning** that evaluates models' ability to discriminate risk type, assess severity, and justify rationales from text.
- Operationalizing a comprehensive risk taxonomy and engineering a high-quality contrastive dataset to stress-test model safety and alignment.

University of Washington

Sep 2020 – Present

Research Lead & Mentor

Seattle, WA

- Recruited and led a 12-person team to generate complex ground-truth datasets for agent benchmarking. Designed consensus protocols to resolve conflicting domain priorities (Data Science vs. ML) to ensure alignment and rigor. **Co-authored paper accepted to EMNLP 2024**
- Advised 50+ interdisciplinary teams through 0-to-1 development of user-centered systems in graduate level project-based classes. Guided the translation of ambiguous user insights into concrete technical specifications, helping teams to evaluate feasibility and manage architectural trade-offs to ensure successful MVP delivery within 10-week constraints.

Microsoft Research

Jun 2025 – Sep 2025

Research Intern

Advisors: Denae Ford Robinson, Edward Cutrell

- Designed a novel structured **user profiling framework** that synthesizes interaction data with social science theory to create robust, interpretable user representations, enabling inference-time personalized steering.
- Engineered a novel ecologically valid evaluation pipeline to generate ground-truth test cases for auditing agent alignment and validated system effectiveness via a robust auditing protocol using **90 days of real user communication data** across 20 participants. **Paper and Patent currently under review.**

Microsoft Research

Jun 2023 – Sep 2023

Research Intern

Advisors: Steven Drucker; Bongshin Lee; Dave Brown

- Led foundational research on using generative AI for end-to-end data analysis and decision making.
- Designed and ran studies with industry experienced data analysts to **evaluate how they understand, trust, and verify AI-generated analyses**, identifying failure modes and verification strategies to inform the design of AI-assisted data analysis tool. **Co-authored paper accepted to CHI 2024**

- Improved the Machine Learning (ML) model (e.g. classification, regression) debugging experience for data scientists by leading research and designing product guided workflows from research insights for TruEra **ML Model Diagnostics**. Identified the top pain points in real debugging practice and prioritized fixes, driving 0 to 1 product exploration during after new funding round and shaped the roadmap.

Selected Publications

▲ Responsible AI ■ Measurement & Benchmarks ★ Agents & LM-powered Systems

Ruoxi Shang, Dan Marshall, Edward Cutrell, Denae Ford Robinson. *Mimetic Alignment with PRISMA: Auditing of AI-inferred Personal Profiles*. *arXiv* (under review). ▲★

Meziah Ruby Cristobal*, Hyeonjeong Byeon*, Tze-Yu Chen*, **Ruoxi Shang***, Donghoon Shin*, Ruican Zhong*, Tony Zhou*, Gary Hsieh. *PaperTok: Exploring the Use of Generative AI for Creating Short-form Videos for Research Communication*. *arXiv* (under review). [* equal contribution] ★

Ruoxi Shang, Ruican Zhong, Yuren Pang, David McDonald, Chirag Shah, Gary Hsieh. *Synthetic Diversity: How Researchers Perceive and Engage with LLM-Generated Diverse Research Feedback*. *arXiv* (under review). ▲■

Ruoxi Shang, Lei Cai, Courtney Cho, Chirag Shah, Gary Hsieh. *Creative Stimuli for Multi-Agent LLM Brainstorming*. *arXiv* (under review). ★

Ruoxi Shang, Keri Mallari, Wei Bin Au Yeong, Ken Yasuhara, Anthony Tang, Gary Hsieh. *Rethinking Teaching Evaluation Reports*. *CSCW 2025*. ▲■

Ruoxi Shang, Gary Hsieh, Chirag Shah. *Trusting your AI Agent Emotionally and Cognitively: Development and Validation of a Semantic Differential Scale for AI Trust*. *AIES 2024*. ▲■

Ken Gu, **Ruoxi Shang**, Ruijen Jiang, Keying Kuang, et al. *BLADE: Benchmarking Language Model Agents for Data-Driven Science*. *EMNLP 2024*. ■★

Ken Gu, **Ruoxi Shang**, Tim Althoff, Chenglong Wang, Steven M. Drucker. *How Do Analysts Understand and Verify AI-Assisted Data Analyses?* *CHI 2024*. ▲★

Ruoxi Shang, Kevin Feng, Chirag Shah. *Lay Users' Needs of Counterfactual Explanations*. *FAccT 2022*. ▲